

Updated on 05/23/2026

Hyeok Yoon

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Experimental Physicist Applying AI to Materials, Devices, and Smart Manufacturing

RESEARCH INTERESTS

Industry: Smart manufacturing and process optimization using machine learning (1 year, on-going).

Academic: Quantum material synthesis and fabrication (13 years).

EDUCATION

Ph.D. Stanford University, Applied Physics (2020).

B.S. KAIST, Physics (2012).

PROFESSIONAL EXPERIENCE

Staff Engineer, Samsung Electronics, Device eXperience 03/2026 – current

- Developing predictive modeling frameworks for material and process properties based on experimental and simulation data.
- Developing core AI technologies applicable to process and material development.

Innocore Postdoctoral Fellow, KAIST 09/2025 – 02/2026

Advisor: Prof. Seunghwa Ryu, Chanho Yang

- Applied physics-guided generative AI models for inverse reconstruction of vortex states in Josephson junction systems.

Postdoctoral Researcher, University of Maryland, College Park 08/2020 – 08/2025

Advisor: Prof. Johnpierre Paglione

- Developing heterojunctions based on spin-triplet topological superconductor uranium ditelluride.
- Developing novel mechanical exfoliation method for non-van der Waals materials.

Postdoctoral Researcher, Stanford University 01/2020 – 07/2020

Graduate research assistant, Stanford University 03/2013 – 01/2020

Advisor: Prof. Harold Y. Hwang

- Developing oxide tunnel junctions and characterizing and analyzing tunneling spectra.
- Thesis: Electronic structure of strontium titanate probed by planar tunneling spectroscopy

Undergraduate research assistant, KAIST 02/2011 – 12/2011

Advisor: Prof. YongKeun Park

SKILLS

Materials Growth	pulsed laser deposition, metal evaporation, atomic layer deposition, magnetron sputtering
Fabrication	lithography, etching, metallization, focused ion beam, exfoliation
Characterization (Structural, optical)	X-ray diffraction, atomic force microscopy, Raman spectroscopy, scanning electron microscopy, spectroscopic ellipsometry
Characterization (Electrical)	- transport, tunneling spectroscopy, electrostatic gating, capacitance, heat capacity, thermal conductivity, mutual inductance - cryogenics (PPMS, MPMS, dilution refrigerators, 3He probe)
Software	Python, MATLAB, Labview, CAD, Igor, Clewin
AI/ML	Machine learning for scientific and process data; Bayesian Optimization, XGBoost, Generative AI and diffusion models, Computer vision and segmentation models (SAM)

PUBLICATIONS

1. **H. Yoon***, S. Baek*, S. Saha, R. E. Butera, J. Paglione, Submicrometer-thick UTe₂ flake achieved by mechanical exfoliation, *Supecond. Sci. Tech.* 39, 045005 (2026) ◇
2. **H. Yoon**, Y. Eo, J. Park, J. A. Horn, R. G. Dorman, S. R. Saha, I. M. Hayes, I. Takeuchi, P. M. R. Brydon, J. Paglione, Probing *p*-wave superconductivity in UTe₂ via point-contact junctions, *npj Quantum Mater.* 9, 91 (2024). ◇
3. **H. Yoon**, A. G. Swartz, S. P. Harvey, H. Inoue, Y. Hikita, Y. Yu, S. B. Chung, S. Raghu and H. Y. Hwang, Low density superconductivity of SrTiO₃ bounded by the adiabatic criterion, arXiv:2106.10802, (2021). (under review; *Proc. Natl. Acad. Sci. U.S.A*) ◇
4. J. Park*, **H. Yoon***, T. Wong, R. Pant, S. Baek, S. Saha, X. Zhang, J. Paglione, S. Lee, I. Takeuchi, Topological YB₆/SmB₆/YB₆ trilayer Josephson junctions, in preparation ◇
5. S. K. Lewin, P. Czajka, C. E. Frank, G. S. Salas, **H. Yoon**, Y. S. Eo, J. Paglione, A. H. Nevidomskyy, J. Singleton, N. P. Butch, High-Field superconducting Halo in UTe₂, *Science* 389, 512 (2025) ◇
6. K. E. Avers, Y. S. Eo, **H. Yoon**, J. A. Horn, S. R. Saha, A. Suarez, P. Zavalij, J. Paglione, Disordered two-dimensional ferromagnetism at the surface of FeSi, *Phys. Rev. B* 110, 134416 (2024). ◇
7. J. Park, J. A. Horn, D. J. Kirsch, R. K. Pant, **H. Yoon**, S. Baek, S. Sarker, A. Mehta, X. Zhang, S. Lee, R. Greene, J. Paglione, I. Takeuchi, Superconducting phase diagram in Bi_xNi_{1-x} thin films: The effects of Bi stoichiometry on superconductivity, *Phys. Rev. Mater.* 8, 074805 (2024). ◇
8. C. E. Frank, S. K. Lewin, G. S. Salas, P. Czajka, I. M. Hayes, **H. Yoon**, T. Metz, J. Paglione, J. Singleton, N. P. Butch, Orphan high field superconductivity in non-superconducting uranium ditelluride, *Nat. Commun.* 15, 3378 (2024). ◇
9. Y. S. Eo, K. Avers, J. A. Horn, **H. Yoon**, S. R. Saha, A. Suarez, M. S. Fuhrer, J. Paglione,

- Extraordinary bulk-insulating behavior in the strongly correlated materials FeSi and FeSb₂, *Appl. Phys. Lett.* 122, 233102 (2023). ◇
10. B. Fauque, C. Collignon, **H. Yoon**, Ravi, X. Lin, I. I. Mazin, H. Y. Hwang, K. Behnia, Electronic band sculpted by oxygen vacancies and indispensable for dilute superconductivity, *Phys. Rev. Research* 5, 033080 (2023). ◇
 11. E. Persky, **H. Yoon**, Y. Xie, H. Y. Hwang, J. Ruhman, B. Kalisky, Electrostatic modulation of lateral carrier density profile in field effect devices with nonlinear dielectrics, *Phys. Rev. B* 107, 195427 (2023). ◇
 12. L. Zhu, H. Liu, M. Somayazulu, Y. Meng, P. A. Gunka, T. B. Shiell, C. Kenney-Benson, S. Chariton, V. B. Prakapenka, **H. Yoon**, J. A. Horn, J. Paglione, R. Hoffmann, R. E. Cohen, T. A. Strobel, Superconductivity in SrB₃C₃ clathrate, *Phys. Rev. Research* 5, 013012 (2023). ◇
 13. R. S. Bisht, M. Mograbi, P. K. Rout, G. Tuvia, **H. Yoon**, A. G. Swartz, H. Y. Hwang, and Y. Dagan, Concomitant appearance of conductivity and superconductivity in (111) LaAlO₃/SrTiO₃ interface with metal capping, *Phys. Rev. Mater.* 6, 044802 (2022). ◇
 14. D. Li, C. Adamo, B. Y. Wang, **H. Yoon**, Z. Chen, S. S. Hong, D. Lu, Y. Cui, Y. Hikita, and H. Y. Hwang, Stabilization of Sr₃Al₂O₆ growth templates for ex situ synthesis of freestanding crystalline oxide membranes, *Nano Lett.* 21, 4454 (2021). ◇
 15. Z. Chen, B. Y. Wang, A. G. Swartz, **H. Yoon**, Y. Hikita, S. Raghu, and H. Y. Hwang, Universal behavior of the bosonic metallic ground state in a two-dimensional superconductor, *npj Quantum Mater.* 6, 15 (2021). ◇
 16. E. Persky, N. Vardi, A. M. R. V. L. Monteiro, T. C. van Thiel, **H. Yoon**, Y. Xie, B. Fauqué, A. D. Caviglia, H. Y. Hwang, K. Behnia, J. Ruhman, and B. Kalisky, Non-Universal current flow near the metal-insulator transition in an oxide interface, *Nat. Commun.* 12, 3311 (2021). ◇
 17. M. Osada, B. Y. Wang, B. H. Goodge, K. Lee, **H. Yoon**, K. Sakuma, D. Li, M. Miura, L. F. Kourkoutis, and H. Y. Hwang, A Superconducting praseodymium nickelate with infinite layer structure, *Nano Lett.* 20, 5735 (2020). ◇
 18. H. Inoue, **H. Yoon**, T. Merz, A. G. Swartz, S. S. Hong, Y. Hikita, and H. Y. Hwang, High-mobility δ -doped field effect transistor, *Appl. Phys. Lett.* 114, 231605 (2019). Editor's pick ◇
 19. Z. Chen, A. G. Swartz, **H. Yoon**, H. Inoue, T. Merz, D. Lu, Y. Xie, H. Yuan, Y. Hikita, S. Raghu, and H. Y. Hwang, Carrier density and disorder tuned superconductor-metal transition in a two-dimensional electron system, *Nat. Commun.* 9, 1 (2018). ◇
 21. A. G. Swartz, A. K. C. Cheung, **H. Yoon**, Z. Chen, Y. Hikita, and H. Y. Hwang, Superconducting tunneling spectroscopy of spin-orbit coupling and orbital depairing in Nb:SrTiO₃, *Phys. Rev. Lett.* 121, 167003 (2018). ◇
 22. K. Kim*, **H. Yoon***, M. Diez-Silva, M. Dao, RR. Dasari, and YK. Park, High-resolution three-dimensional imaging of red blood cells parasitized by plasmodium falciparum and in situ hemozoin crystals using optical diffraction tomography, *J. of Biomedical Optics* 19(1), 011005 (2013). ◇

PRESENTATIONS

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1. Poster, International Workshop on Oxide Electronics (iWOE-21), Bolton Landing, NY (Sep. 2014)
 2. Poster, Physics and Chemistry of Surfaces and Interfaces (PCSI), Palm Springs, CA (Jan. 2016)
 3. Oral, American Physics Society (APS) March Meeting, Baltimore, MD (Mar. 2016)
 4. Poster, Canadian Institute for Advanced Research (CIFAR) Quantum Materials Meetings, Vancouver, BC (Apr. 2017)
 5. Poster, International Workshop on Oxide Electronics (iWOE-24), Chicago, IL (Sep. 2017)
 6. Oral, International Conference on Magnetism (ICM), San Francisco, CA (Jul. 2018)
 7. Oral, American Physics Society (APS) March Meeting, Boston, MA (Mar. 2019)
 8. Oral and Poster, Strongly Correlated Electron Systems (SCES), Okayama, Japan (Sep. 2019)

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9. Oral, Emergent Phenomena in Quantum Systems Initiative (EPiQS) Postdoctoral Symposium, Beverly, MA (May. 2022)
10. Oral (Invited), Gordon Research Seminar (GRS): Correlated Electron Systems, South Hadley, MA (June. 2022)
Poster, Gordon Research Conference (GRC): Correlated Electron Systems, South Hadley, MA (June. 2022)
11. Poster, Materials and Mechanisms of Superconductivity (M2S), Vancouver, BC (July. 2022)
12. Oral, Strongly Correlated Electron Systems (SCES), Incheon, Korea (Aug. 2023)
13. Oral (Invited), Canadian Institute for Advanced Research (CIFAR), New York, NY (Nov. 2023)
14. Oral, American Physics Society (APS) March Meeting, Minneapolis, MN (Mar. 2024)
15. Poster, International Conference on Advanced Materials and Devices (ICAMD), Busan, Korea (Dec. 2025)
16. Oral (Invited), Korean Dielectric Society (KDS) Winter Conference, Muju, Korea (Feb. 2026)

SEMINARS

1. Center for Correlated Electron Systems, Institute of Basic Science (IBS), Seoul, Korea (Aug. 2019)
2. Department of Physics, Korea Institute of Advanced Science and Technology (KAIST), Daejeon, Korea (Aug. 2019)
3. Department of Physics, Chung-Ang University, Seoul, Korea (Sep. 2024)
4. Korea Atomic Energy Research Institute, Daejeon, Korea (Sep. 2024)
5. Department of Physics, Korea Institute of Advanced Science and Technology (KAIST), Daejeon, Korea (Sep. 2025)
6. Department of Nano-semiconductor Engineering, Ulsan University, Daejeon, Korea (Dec. 2025)

AWARDS AND HONORS

- ICAMD Young Researcher Award (2025)
- Accelnet Quantum Materials Exchange Award (2024, 2023)
- Kwanjeong Graduate Fellowship (09/2012 – 08/2016)
- National Science and Engineering Undergraduate Scholarship, Korea Student Aid Foundation, (2008-2011)
- Undergraduate Student Scholarship, Korea Foundation for Advanced Studies, (2009-2011)

OUTREACH AND EXPERIENCES

- I served as a reviewer for funding proposals in Gordon and Betty Moore Foundation and Air Force Office of Scientific Research.
- I served as a reviewer in Physical Review X and Physical Review B.
- I served as a synthesis module runner for the workshop on Fundamentals of Quantum Materials School 2025.
- I presented a career talk for undergraduate students at Physics department in KAIST